

Multiple Choice (10 marks)

1. What pressure would 131g of xenon gas in a vessel of volume 1.0dm^3 exert at 25°C if it behaved as a van der Waals gas?
 - (a) 22atm
 - (b) 20atm
 - (c) 15atm
 - (d) 30atm
 - (e) 34atm
2. Determine the equilibrium constant for the reaction $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ the equilibrium concentrations are: H_2 , 0.9 moles/liter; I_2 , 0.4 mole/liter; HI, 0.6 mole/liter.
 - (a) 0.9
 - (b) 1.0
 - (c) 2.5
 - (d) 0.3
 - (e) 0.5
3. The temperature of the fireball in a thermonuclear explosion can reach temperatures of approximately 10^7 K. What value of λ_{max} does this correspond to?
 - (a) 2×10^{-10} m
 - (b) 1×10^{-10} m
 - (c) 7×10^{-10} m
 - (d) 6×10^{-10} m
 - (e) 4×10^{-10} m
4. The density of a gaseous compound was found to be 1.23kgm^{-3} at 330K and 20kPa. What is the molar mass of the compound?
 - (a) 169 g mol^{-1}
 - (b) 123 g mol^{-1}
 - (c) 200 g mol^{-1}
 - (d) 150 g mol^{-1}
 - (e) 190 g mol^{-1}
5. A liquid element that is a dark-colored, nonconducting substance at room temperature is

- (a) lead
- (b) cesium
- (c) bromine
- (d) francium
- (e) rubidium

True / False (10 marks)

Answer True or False

6. True or False: The height of the mercury column corresponding to a 600 mm water column is approximately 44.1 mm Hg.
7. True or False: The most massive subatomic particle is the proton, which has a greater mass than the positron.
8. True or False: The logarithm of 0.0364 is -1.5000, indicating a common misconception about logarithmic values.
9. True or False: The original energy level of the electron for this transition in the Lyman series is 4.
10. True or False: To completely combust 1000 cubic feet of butane, 6500 cubic feet of oxygen are required based on stoichiometric calculations.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the volume of concentrated sulfuric acid needed to achieve a mass of 74.0 g, given its density is 1.85 g/ml. Discuss your approach and reasoning in detail.
12. A chemist dropped a 200 g object into water, displacing 60 ml, and then into oil, displacing 50 g. Calculate and discuss the densities of both the object and the oil, explaining the significance of your findings.

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